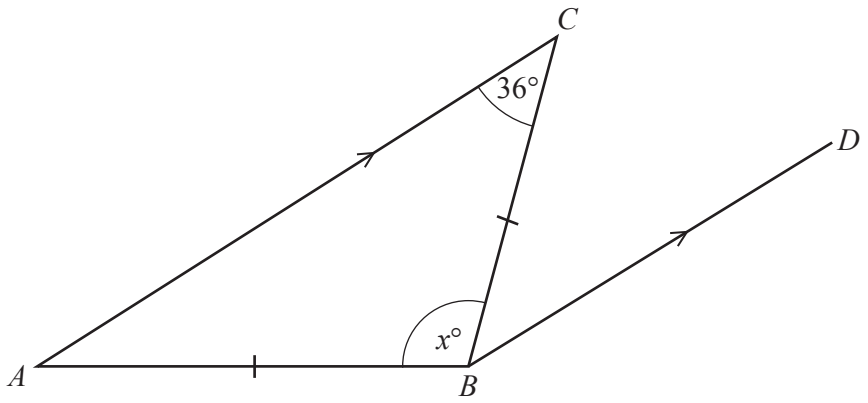
0580_s21_qp_32_Q8

8 (a)



NOT TO SCALE

The diagram shows a triangle *ABC* and a line *BD*.
AB = *BC* and *AC* is parallel to *BD*.

(i) Angle *ACB* = 36°.

Write down the mathematical name for this type of angle.

..... [1]

(ii) Write down the mathematical name for triangle *ABC*.

..... [1]

(iii) Work out the value of *x*.

x = ... [2]

(iv) Find angle *CBD*.
Give a geometrical reason for your answer.

Angle *CBD* = ... because .....

..... [2]

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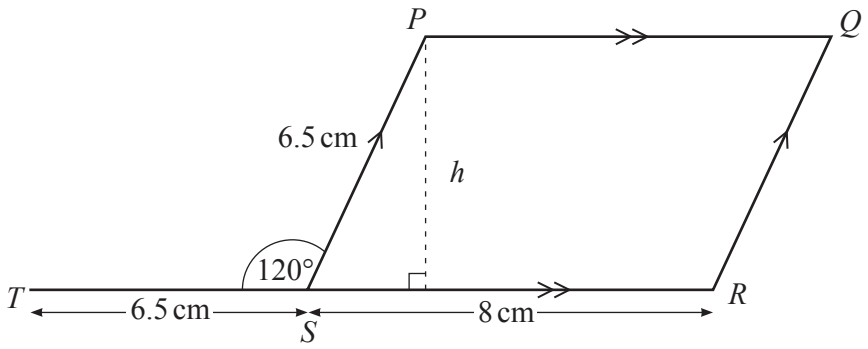
(b)



NOT TO SCALE

The diagram shows a quadrilateral, *PQRS*.
PQ is parallel to *SR* and *SP* is parallel to *RQ*.
TSR is a straight line.
SR = 8 cm, *PS* = *ST* = 6.5 cm and angle *PST* = 120°.

(i) Write down the mathematical name of quadrilateral *PQRS*.

..... [1]

(ii) Work out the perimeter of quadrilateral *PQRS*.

..... cm [1]

(iii) Find angle *PSR*.
Give a reason for your answer.

Angle *PSR* = ... because .....

..... [2]

(iv) *PS* and *ST* are two sides of a regular polygon.

Work out the number of sides of this regular polygon.

..... [1]

(v) Show that the height, *h*, of the quadrilateral *PQRS* is 5.63 cm, correct to 2 decimal places.

[2]

(vi) Work out the area of quadrilateral *PQRS*.

..... cm^2 [2]